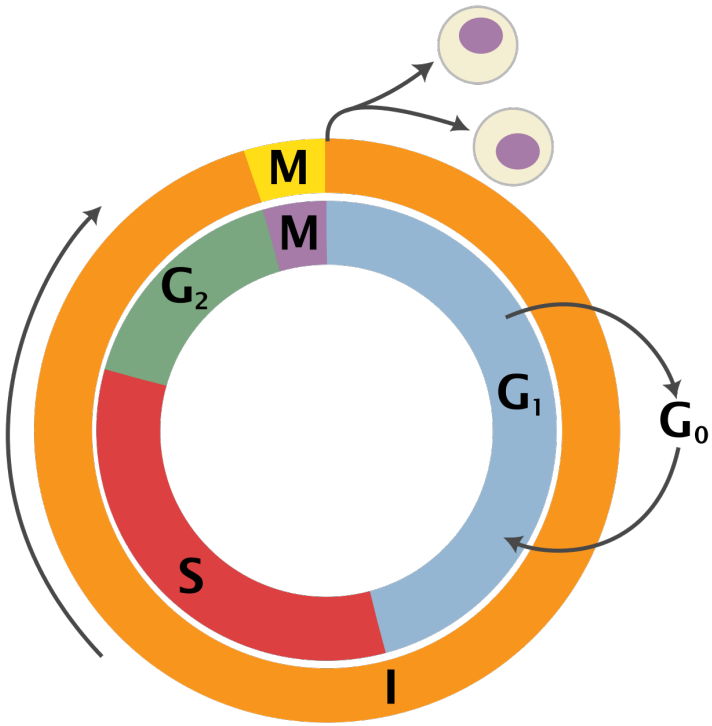
Computational Biology

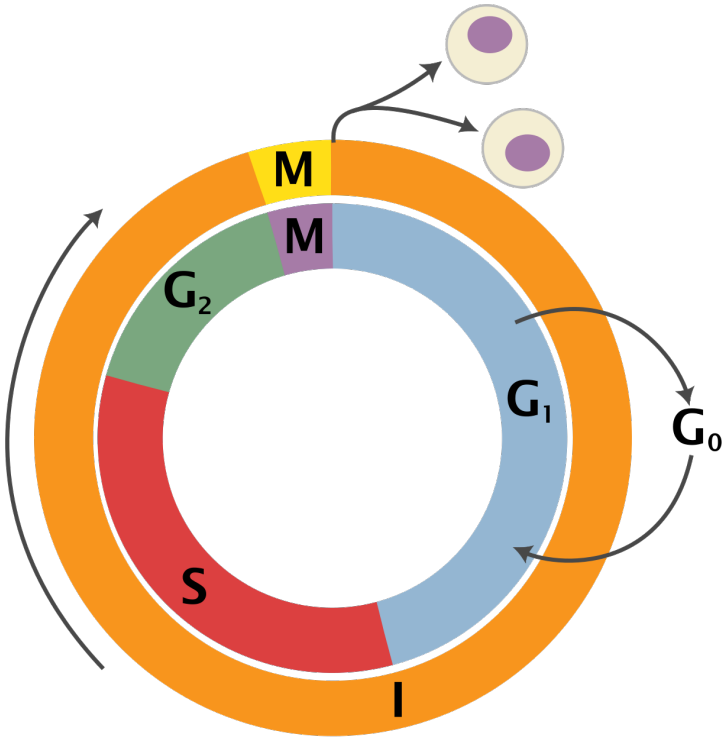
Lecture 20: *Introduction to Lab 1*



Chemical Equations, Stochasticity, Gillespie Algorithm

Outline

1. **The Tutorials**
2. Lab 1



The Labs/Tutorials

- There are four tutorials
- The first tutorial suggests some videos for you to look at—do this in your own time
- The first and later tutorials describes works you need to do
- The coursework is to be done individually
- Each tutorial is worth 7.5%

Hand-In

- Use the hand-in system to submit your work as a PDF (not as a jupyter notebook)
- Hand-in dates are

1. Due 4pm on Thursday 18th February 2022
2. Due 4pm on Thursday 24th March 2022
3. Due 4pm on Thursday 5th May 2022
4. Due 4pm on Thursday 19th May 2022

Help

- We will run this session every week
- We will discuss the labs and broad issues that might help everyone
- We will then split into channels (hence you groups)—I intended to split you into groups of 5, but because of the limitations of Teams I have put two groups together
- If you are not yet in a group the just join a channel
- A team of four of us will go around the 10 channels

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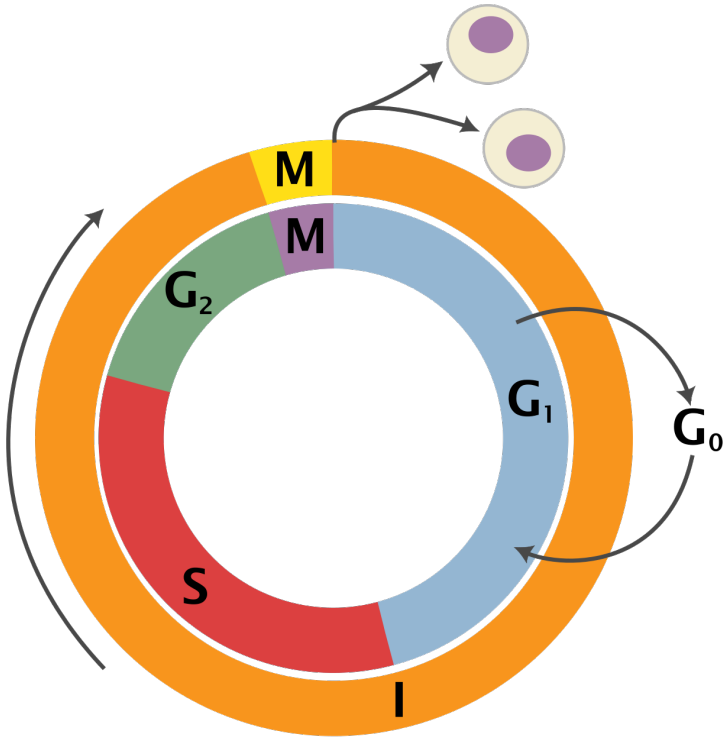
2. **Lab 1**



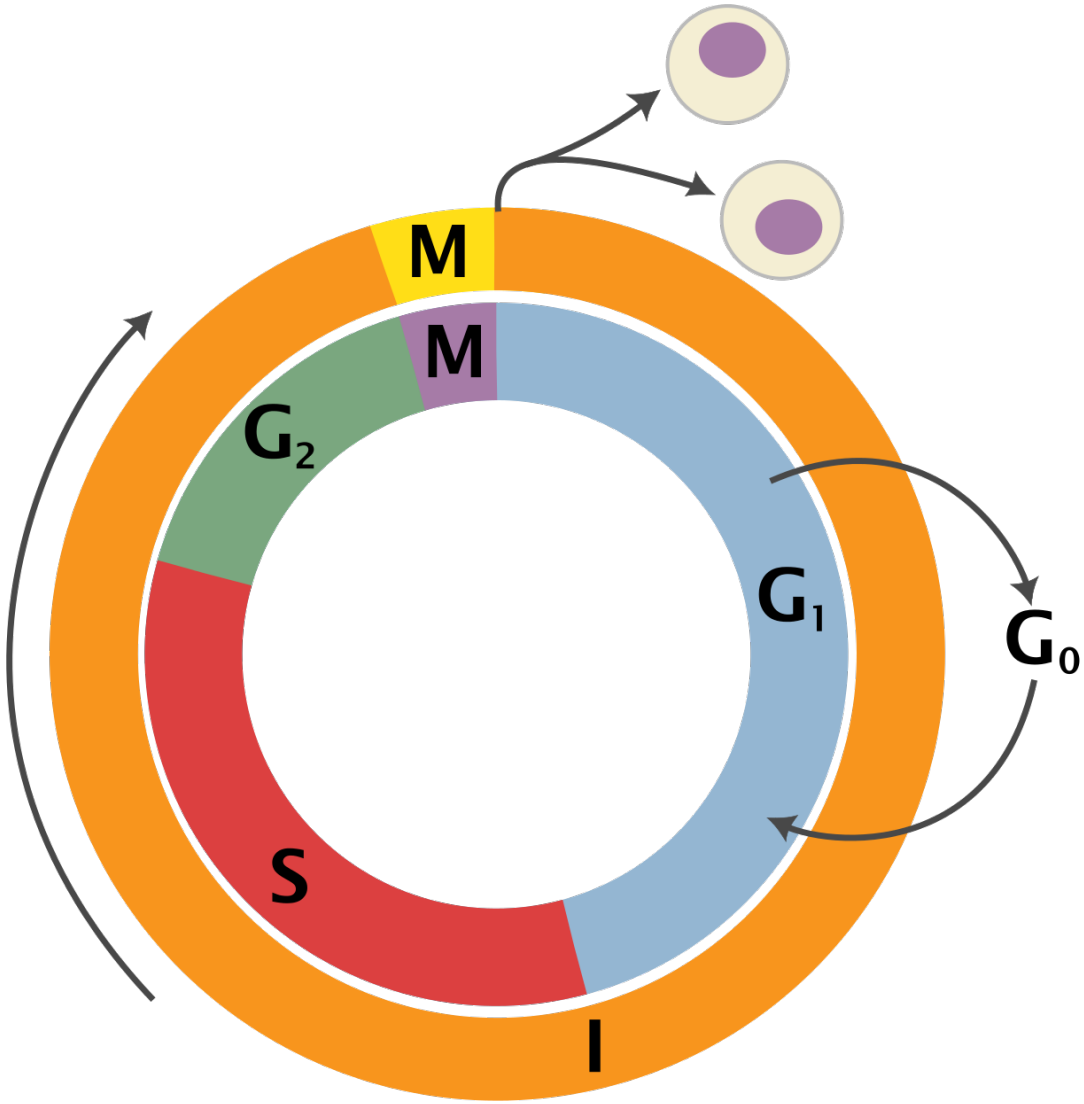
Data Exploration

- The first lab looks at biological data exploration
- The data consists of RNA measurements and raw protein measurements
- RNA measurements acts as a proxy for protein concentration (it is cheap to measure)
- The protein measurements are real protein concentration but much more expensive to do

The Data

- The data is from humans
- It is measure at three points in the cell-cycle
- That is, cells regularly grow and divide
- These cells were cleverly synchronised so we could measure the protein concentration at different points in the cell cycle

The Cell Cycle



Python and Jupyter Notebook

- We provide python code to do the basic data exploration
- You are welcome to use this or to ignore it
- To get full marks we are expecting you to have gone beyond what we give you
- We provide a jupyter notebook. You don't need to use this, but it is a reasonably nice way to visualise what you are doing
- Please don't submit the raw notebook—you can turn this into PDF

Pandas

- Pandas is a python package that dominates modern data science
- Other package are available—but they are nearly all based on pandas
- Pandas has two data structures: `Series` and `DataFrame`
- DataFrame is a glorified table with labels for the columns and rows
- It has a lot of functionality built around it for rapid data exploration

Matplotlib

- Matplotlib is a visualisation library
- It was designed to look like MATLAB plot functions and is consequently very ugly
- It has a lot of functionality, but isn't particularly natural to use
- Seaborn is build on top of matplotlib and produces slightly prettier graphs (but is still not easy to use)
- Sometime you need to write `> %matplotlib inline` to get the graphs to show